



Offset+

Offset surfaces, or curves, toward the side a reference is on -- surfaces of opposite orientation all move the same way, and curves offset in the frame you choose.

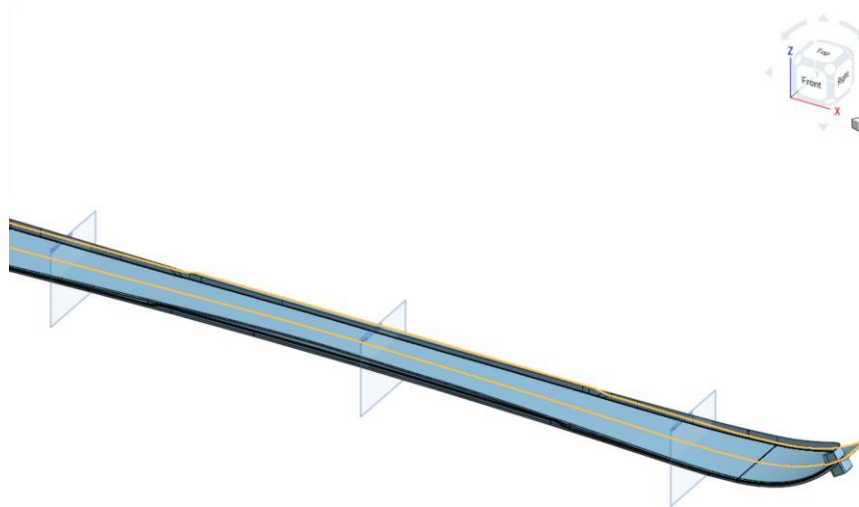


What it does

The built-in offset surface goes along the normal, and the built-in offset curve works in a sketch plane with a flip. Offset+ offsets each surface body toward the reference's side of it, and offsets 3D curves in a transport, surface-normal or plane frame -- toward the reference too.

Use it when

- Surfaces from different sources face different ways
- A 3D edge chain needs a constant offset (along a surface, lifted off it, or in plan)
- Corners of the chain must be rounded or trimmed cleanly



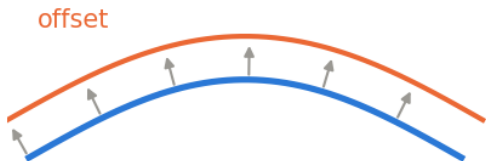
Offset+ open on the ski topsheet: offset 2 mm toward the plane 'Below ski'.



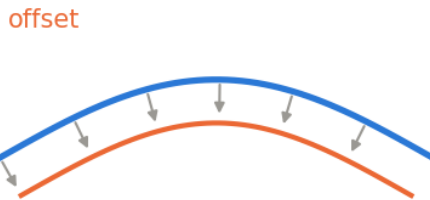
Why a reference instead of a flip

A flip names a side relative to a normal; a reference names it in the model

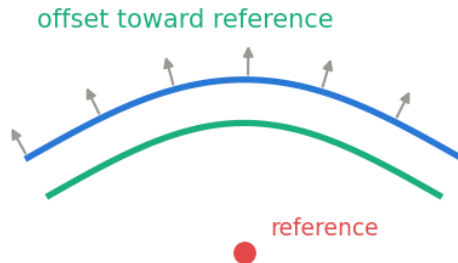
Built-in, normal up:
'offset along normal' goes up



Same model after an upstream change
flips the normal: now it goes DOWN



Offset+ 'toward reference':
goes toward the point, either way



A flip is about a normal

'Front', 'opposite direction', 'Thickness 1'
all mean: relative to the normal -- and
normals change freely upstream.

A reference is about the model

Pick geometry on the side you mean. The
choice survives any change of
orientation.

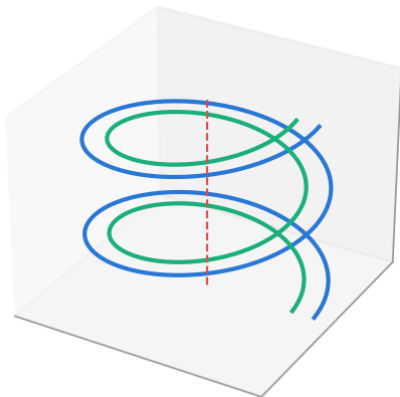
Drop-in

With no reference, the feature behaves
exactly like the built-in.



Curve frames and corners

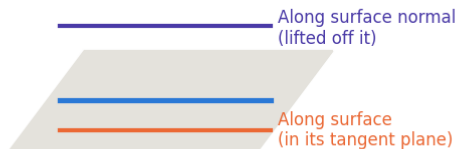
Transport: toward the axis all the way
(C7: 10 mm inside a r50 helix)



Transport frame

The direction is carried along the chain without twisting -- a helix offset 'toward the axis' stays inside (test C7).

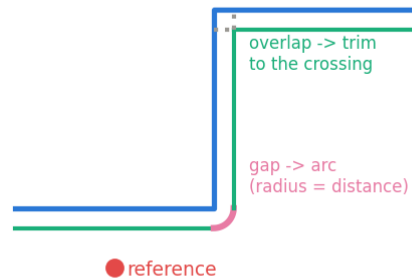
Surface-normal frame: two directions



Surface normal / plane

Off a surface, along it, or in the plane perpendicular to a direction (plan view).

Corners (C8: 1 rounded + 1 trimmed, length $280 + 5 \pi = 295.708$)



Corners

A gap on the outside gets an arc (radius = the distance); an overlap on the inside is trimmed to the crossing (test C8).



The dialog: surfaces

Offset+ topsheet 2 mm to...  

1 **Surface** Curve

2 Faces and surfaces to offset
Filletted_Top_Surf 

3 Distance 2 mm

4 Side reference
Below ski (z -50)   

6 > Debug

#	Parameter	What it means
1	Offset type	SURFACE: offset surface bodies. CURVE: offset edges or wires (next slides).
2	Faces and surfaces to offset	Each body is offset toward the reference's side of THAT body.
3	Distance	Offset distance.
4	Side reference	Geometry on the side to offset toward. Empty = the arrow alone (a plain flip).
5	Offset toward reference	The arrow. On: toward the reference. Off: away from it.
6	Debug	Print details: which way each body or chain went, corners and fits.



The dialog: curves, plane frame

Offset+ edge chain 5 mm... ✓ ✗

1 Surface Curve

2 Curves and edges to offset
Edge of Derived 1 ✕

3 Plane ▼

4 Plane normal

5 Distance 5 mm

6 Side reference ⌚ ↗
Front plane ✕

8 > Fit
> Debug

Final

#	Parameter	What it means
1	Offset type	CURVE.
2	Curves and edges to offset	Connected edges form one chain; each chain becomes one wire.
3	Frame	Transport frame (default) / Surface normal / Plane.
4	Plane normal	Plane frame: the offset lies in planes perpendicular to this. Empty = world Z (a plan-view offset).
5	Distance	Offset distance.
6	Side reference	Here the Front plane: every chain moves toward it.
7	Offset toward reference	The arrow: toward / away.
8	Fit	Samples per edge (24), Max station spacing (5 mm), Fit tolerance (0.001 mm).



The dialog: curves, surface-normal frame

Offset+ edge chain 5 mm... ✓ ✗

Surface Curve

Curves and edges to offset
Edge of Derived 1 ✕

1 Surface normal ▼

2 Surface
Filletted_Top_Surf ✕

3 Direction Along surface ▼

Distance 5 mm

Side reference ⌚ 📍
Front plane ✕

> Fit

> Debug

Final

#	Parameter	What it means
1	Frame	Surface normal: the direction comes from a surface at each point.
2	Surface	The surface whose normals define the offset (its orientation does not matter: normals are kept continuous).
3	Direction	Along surface normal: lift the curve off the surface. Along surface: stay in it, perpendicular to the curve.

On the topsheet edge chain all three frames land at y 70 (5 mm in): the topsheet is flat across its width there. They differ on twisted chains and curved surfaces (tests C5-C7).



Example: surface offset

Offset+ surface: topsheet offset 2 mm toward the plane 'Below ski' (section at x 885)



topsheet (faces down, into the ski)
offset 2 mm toward 'Below ski'

Topsheet offset 2 mm toward 'Below ski' (topsheet_surf studio)

The topsheet faces down, into the ski. Offset+ reads which side of the topsheet 'Below ski' is on and offsets that way -- if the topsheet's normal flipped upstream, the offset would still go down. The offset surface keeps the topsheet's orientation (still facing down).



Outputs and messages

Published for Extract variables

Key	What it holds
output	The offset surfaces or wires.
startVertex / endVertex	Curves: the ends of each wire (start = the end at the source's start).
startEdge / endEdge	Curves: the edge at each end.
cornerArcs	Curves: the arcs that round the outside corners.
boundaryEdges	Surfaces: the open boundary edges of the offset surfaces.

Messages

	Message / what to do
Info	Corners: N rounded, M trimmed. The corner counts are reported here only (not published).
Warning	N corner(s) could not be closed ... The wire has a gap at each. A corner sharper than the offset allows, or skew 3D pieces. Reduce the distance or split the chain there.
Error	The side reference is not to either side of chain k in this frame The reference lies along the curve or on the surface. Move it off to one side.
Error	The curve runs along the plane / surface normal No offset direction exists there in that frame.



Tips and limits



Use a local reference

A mate connector or vertex near the geometry, on the side meant.



Transport for 3D chains

It never flips at an inflection and never twists, so the offset stays on the side it started.



Outputs keep their input's orientation

The reference picks the side, not the normal. Built-ins downstream that follow the normal (Thicken, Offset surface, Move face) need Orient to reference first -- or use the reference-side feature instead.



Sharp corners and large distances

Where the offsets neither separate nor cross cleanly the corner is left open, with a warning.



Where it lives

Document	Reference_Side_Features (22764764a00a7f607dbc1c4d)
Tab	offset_plus (uses reference_side_utils)
Tests	Reference side tests studio: O1-O2 (surfaces), C1-C8 (curves) - devtools/onshape/check_reference_side_tests.py (34/34)
Examples	topsheet_surf studio (the user's derive of Design_Master topsheet and sidewall bodies)
Explainer	docs/explainers/reference_side/reference_side_explained.md, section 2.3
Related	Split+, Mutual Trim+, Offset+, Thicken+, Orient to reference (same reference rule); Extract variables